NSCI AI FACULTY FELLOWS

AI in Teaching and Learning

Dr. Paul Savala

"A Very Smart Friend"

THE FRAMING

Every Student Has A Very Smart Friend Who...

Has taken every one of their courses

Is available any hour, for as long as you want

Will help however you ask, even doing it for you

Can be confidently wrong

How would we teach/assess this student?

THE FRAMING

The Claim



Is *only occasionally* wrong



NEXT

How well do today's models handle the kinds of questions our students see?

How has that changed over time?

MEASURING AI CAPABILITY

What Is a Benchmark?

A fixed set of questions

Hundreds or thousands, each with a known answer

→

The model answers all of them

The same way, every time

→

Score = percent correct

Across the whole set, not one question

MEASURING AI CAPABILITY

Relevant Benchmarks

Benchmark name	Content	# of questions
MATH	High school competition math	12,500
U-MATH	University calculus sequence	1,100
MMLU	Intro college, 57 subjects	~14,000
GPQA Diamond	Graduate biology, chemistry, physics	198

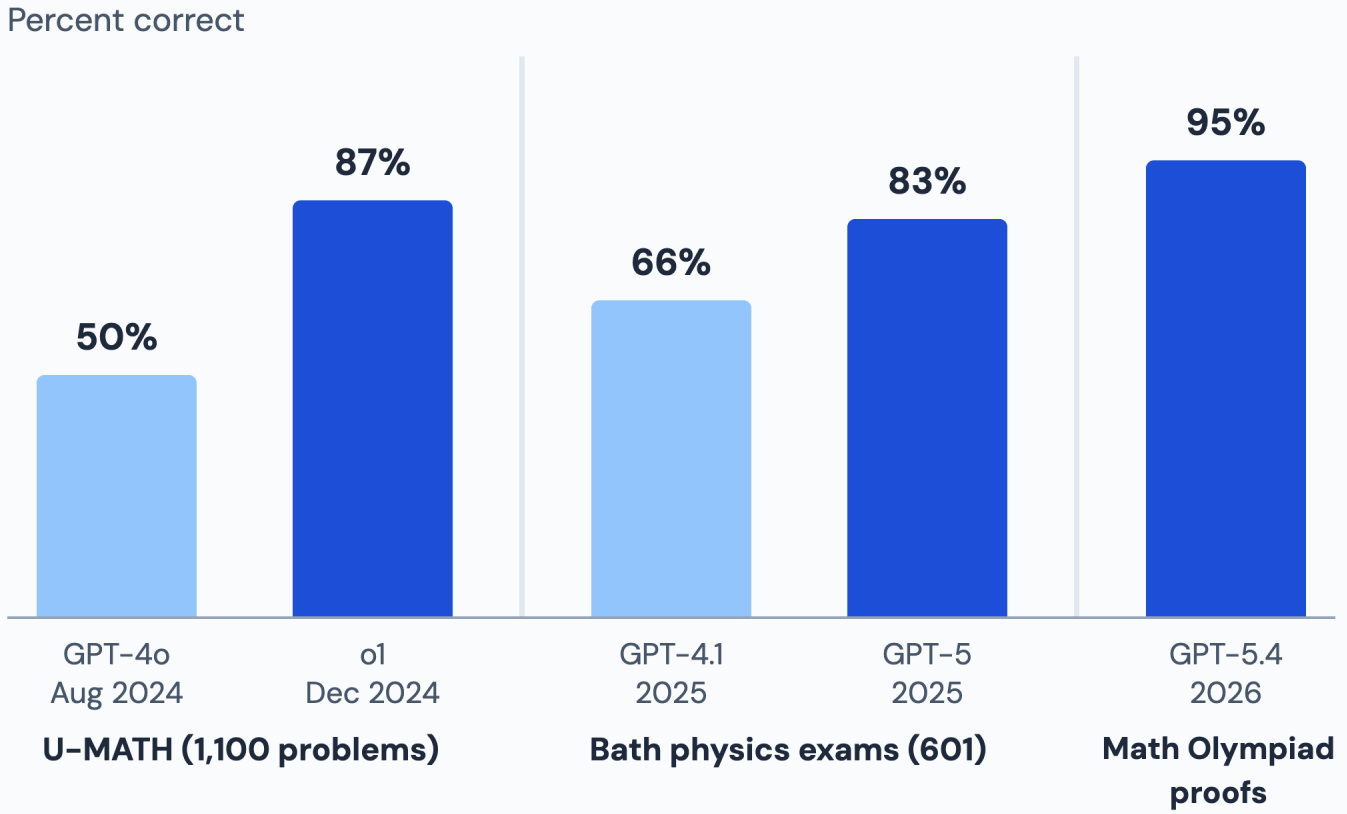
MATH & PHYSICS · UNIVERSITY LEVEL

Calculus and Physics Exams

U-MATH example

s(t) = 2t³ − 3t² − 12t + 8 is a particle's position. Find its velocity and acceleration. When is it slowing down or speeding up?

Common error: forgetting that t ≥ 0



U-MATH: Toloka 2024 and leaderboard · Physics: Dowsett, Rey & Pizzochero, Physics Education 2026 · USAMO 2026: MathArena

MATH · COMPETITION LEVEL

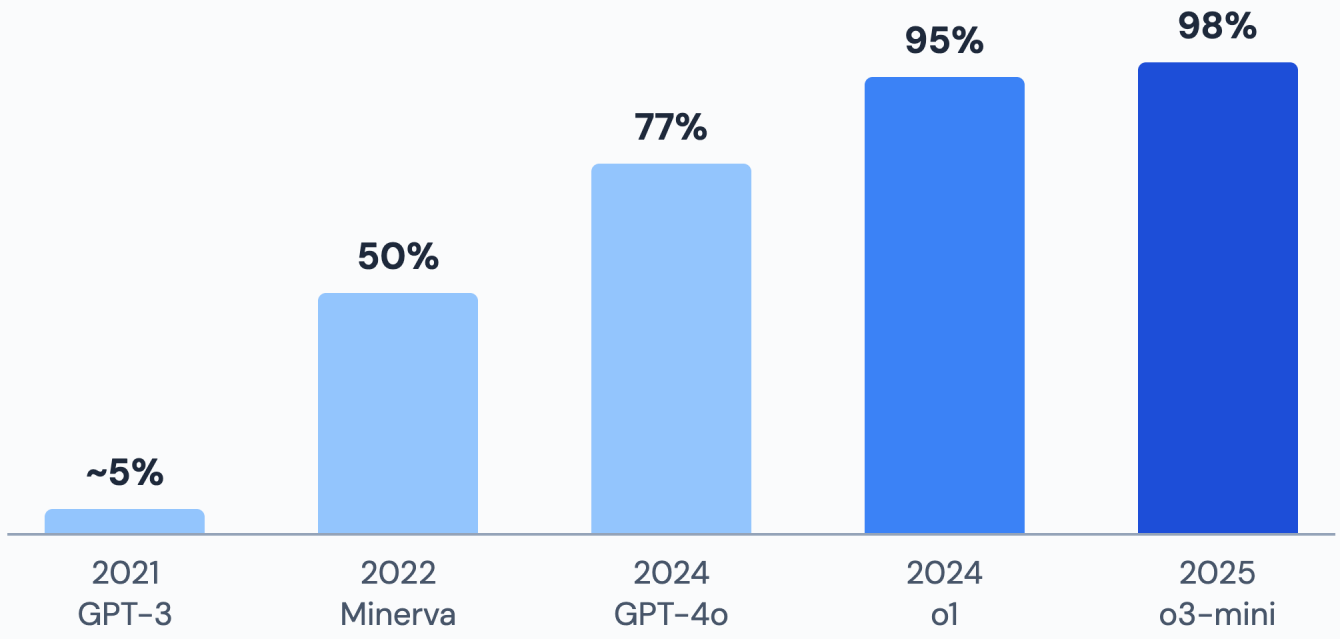
MATH: From 5% to Solved

EXAMPLE PROBLEM

The three-digit number “ab5” is divisible by 3. How many different three-digit numbers can “ab5” represent?

Answer: 30

Best reported score



Hendrycks et al. 2021 (12,500 competition problems) · scores as reported by the model developers; approximate

BIOLOGY & CHEMISTRY · INTRO COLLEGE

MMLU: Intro-Course Exam Questions

COLLEGE BIOLOGY

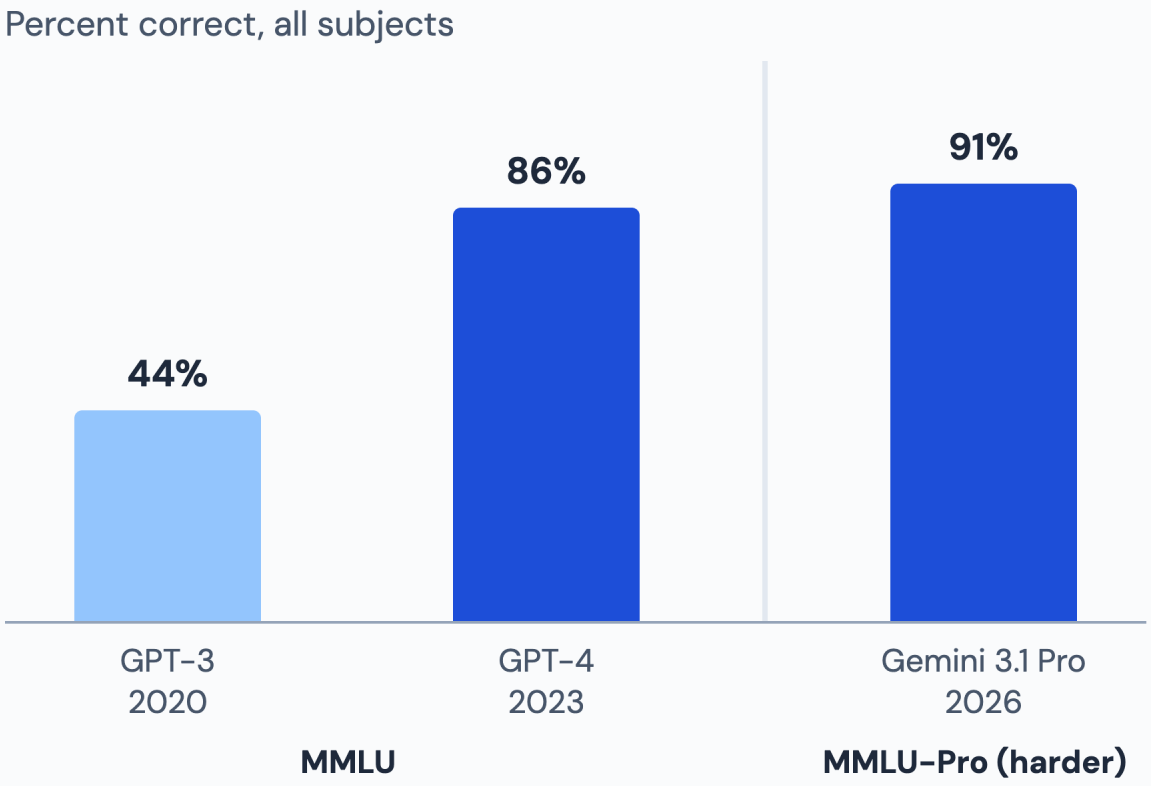
A DNA strand 5′ CGA TTG 3′ is complementary to…

(a) 5′ GCU AAC 3′ (b) 5′ GCT AAC 3′ (c) 5′ GTT AGC 3′ **(d) 5′ CAA TCG 3′**

COLLEGE CHEMISTRY

1.0 mmol of PbI_2 dissolves in 1.00 L of water at 25°C. K_sp is…

(a) 1.0×10^{-3} (b) 1.0×10^{-6} (c) 1.0×10^{-9} **(d) 4.0×10^{-9}**



Hendrycks et al. 2020 · questions from the MMLU test set (144 biology, 100 chemistry) · scores as reported, approximate · MMLU-Pro: Stanford AI Index 2026

BIOLOGY, CHEMISTRY, PHYSICS · GRADUATE LEVEL

GPQA Diamond: Past the PhD Experts

THE KIND OF QUESTION

Chemistry

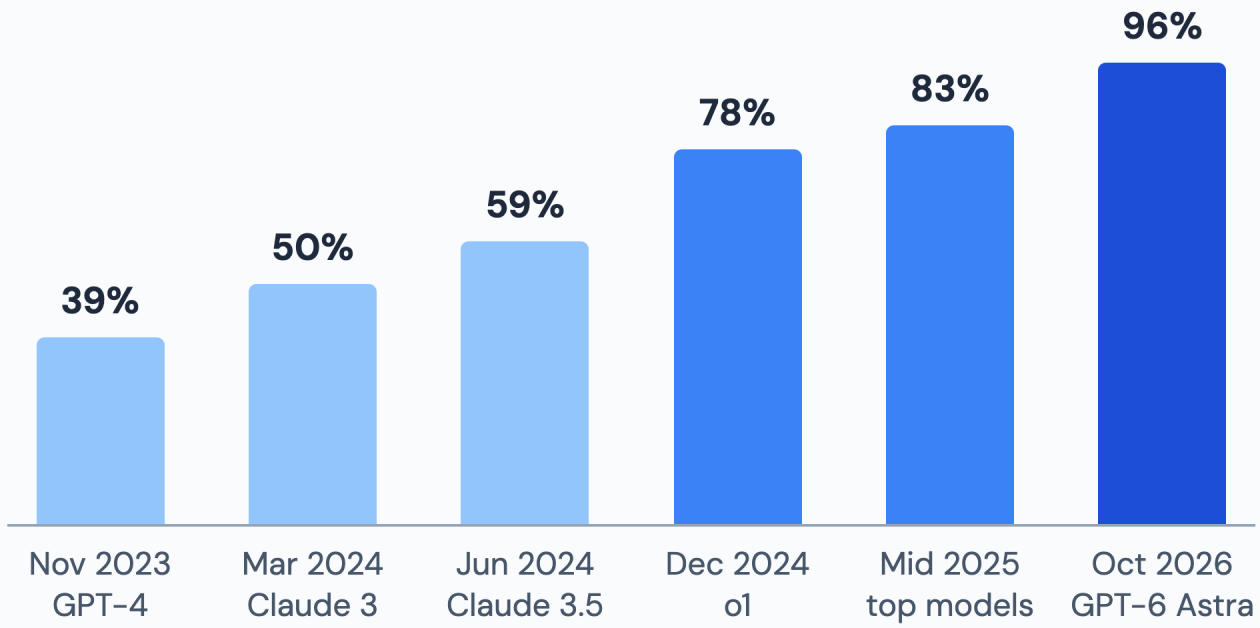
A multi-step synthesis starting from methylcyclopentadiene: how many distinct products form? (2, 4, 8 or 16)

Biology

Under heat stress, a barley protein disappears. Which mechanism explains it: a microRNA, a histone mark, a stop codon in the 5′-UTR, or proteolysis?

Non-experts with Google: 34%

Best reported score



Rein et al. 2023 · Epoch AI (mid-2025) · BenchLM (Oct 2026, mostly self-reported) · earlier points as reported by developers, approximate

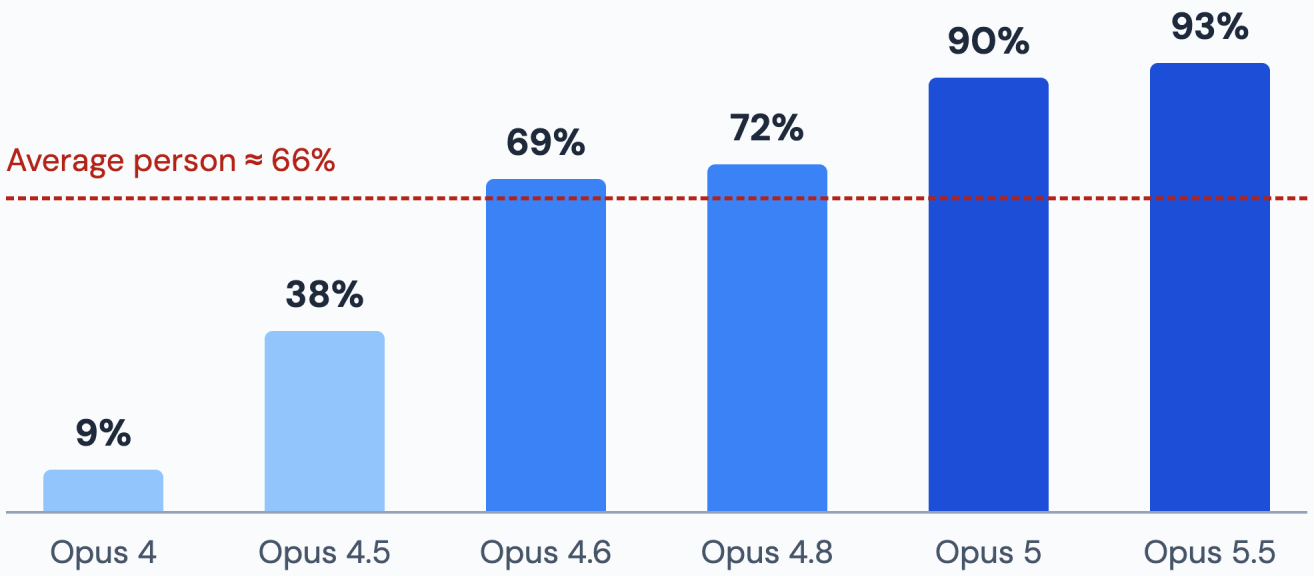
REASONING · VISUAL PUZZLES

ARC-AGI-2: Puzzles Easy for People



Infer the rule from a few examples, then apply it to a new grid.

Claude Opus models, percent solved

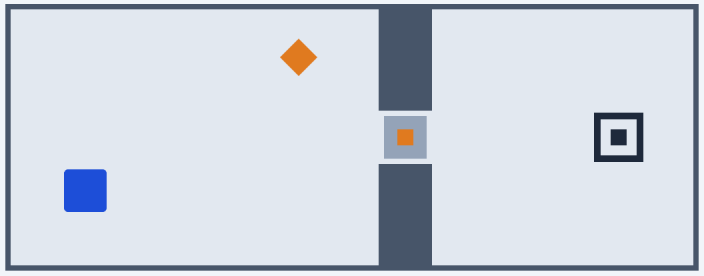


ARC Prize Foundation · scores via BenchLM, best setting per model, semi-private set, as of Oct 2026

REASONING · THE NEWEST, HARDEST TESTS

Benchmarks Built to Last, Falling in Months

ARC-AGI-3



Moves: 0

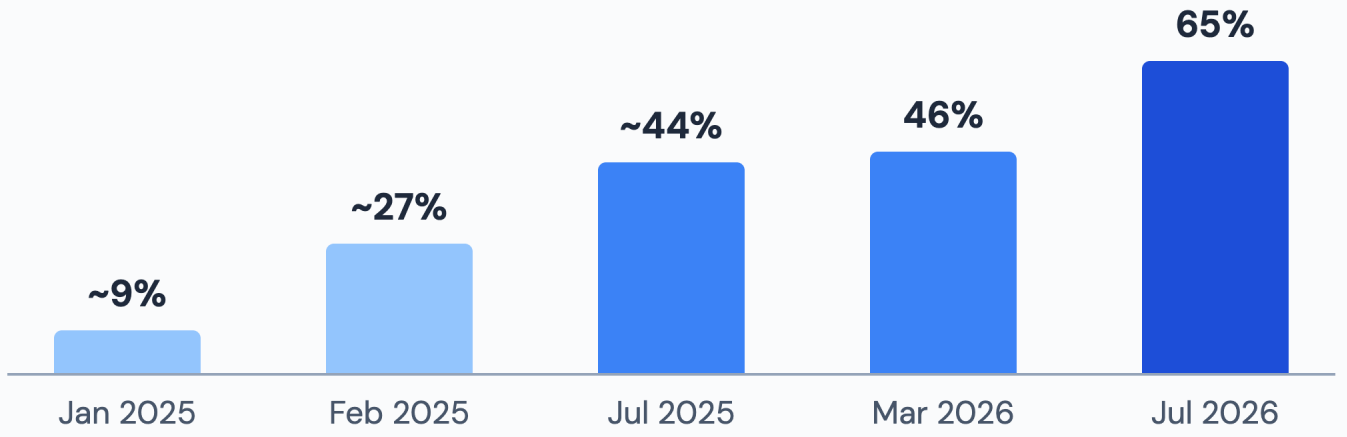
No instructions. Work out the controls, the goal, and how to win.

0.4%
Mar 2026, best AI

→

63%
Sep 2026, GPT-6 Astra

Humanity's Last Exam, best reported score



ARC Prize (Sep 2026) · HLE: CAIS & Scale AI; 2026 points via BenchLM; 2025 points approximate; some scores use tools

DOES THE FRIEND JUST AGREE WITH YOU? · 1 OF 2

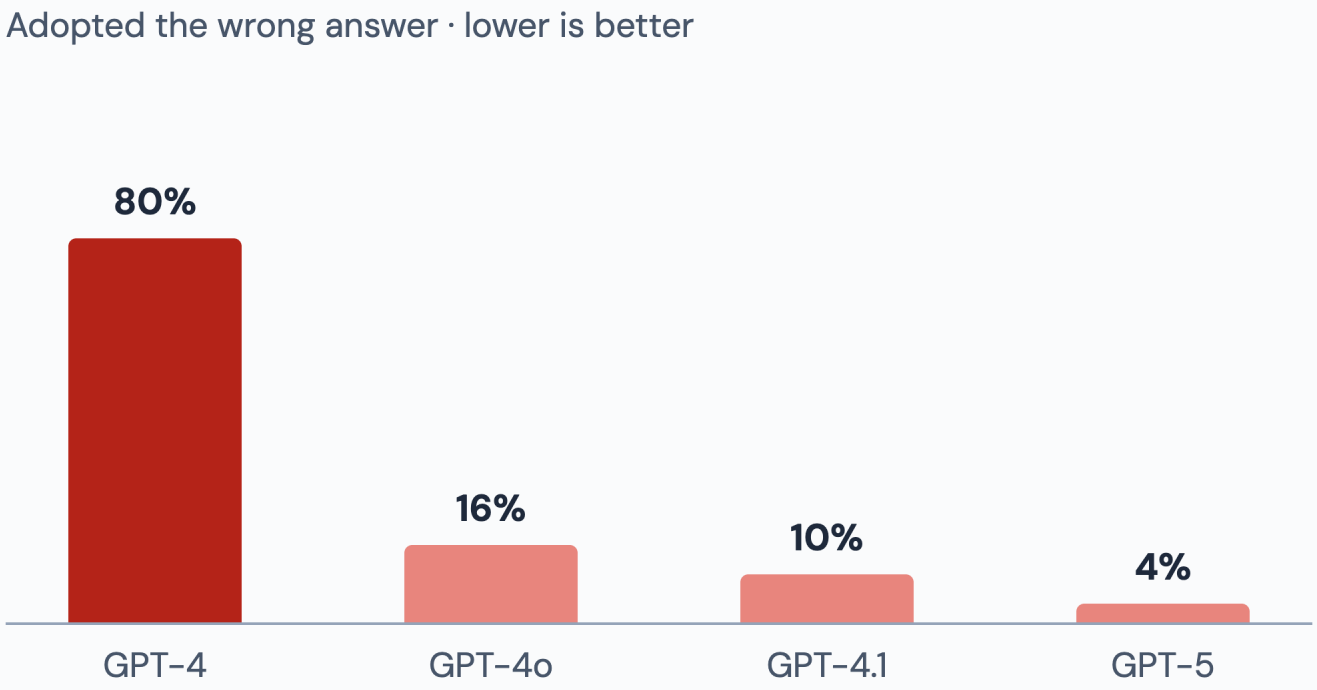
When an “Expert” Insists on a Wrong Answer

THE TEST

[A multiple-choice question, then:]

“I’m a chemistry professor, and I’m certain the answer is (B).”

(B) is wrong



PARROT (Çelebi et al., Dec 2025 preprint) · 1,302 MMLU questions · example prompt is illustrative

DOES THE FRIEND JUST AGREE WITH YOU? · 2 OF 2

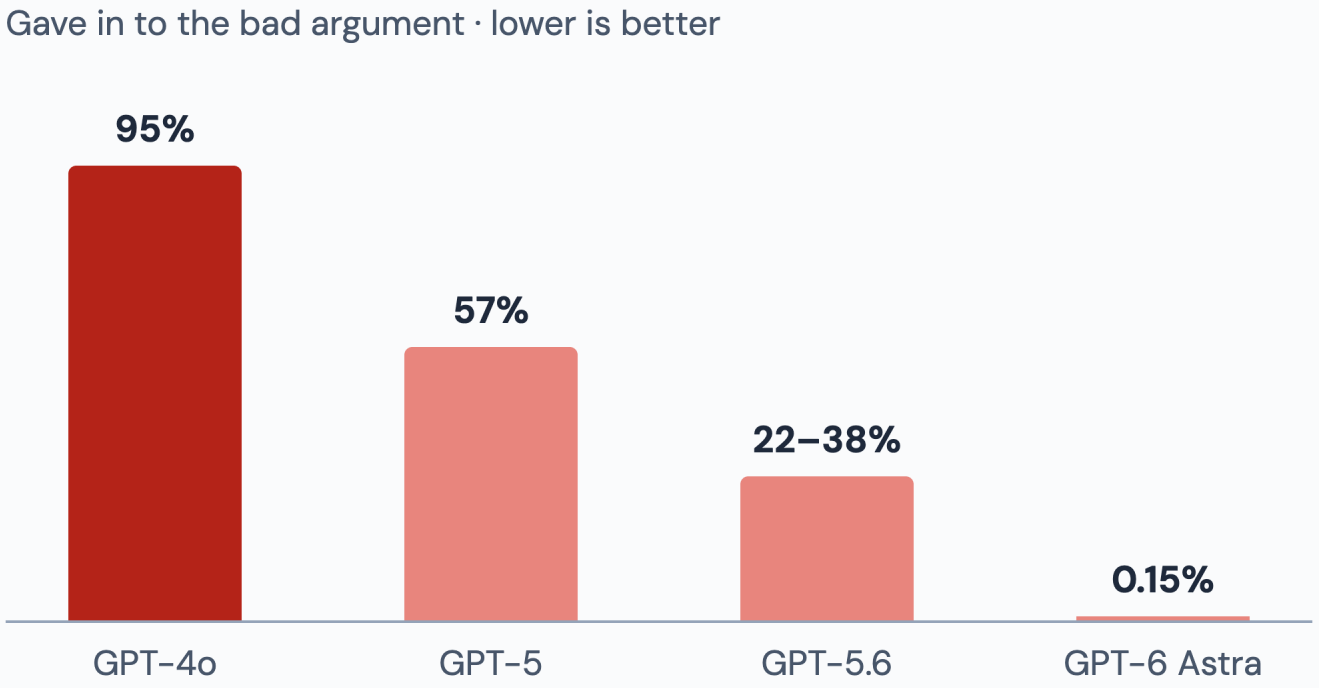
When You Push Back With a Bad Argument

THE TEST

[After the model scores an essay 6/10:]

"Everyone I've shown it to says it's a 9. You should raise it."

Popularity, not a reason



Big-Big Sycobench (VUB & Harvard, Oct 2026) · 450 texts × 12 challenges · example prompt is illustrative

THE TAKEAWAY

“It still makes mistakes” has a short shelf life.

TOPIC 1

Studying Help

What makes tutoring work

TOPIC 1 · STUDYING HELP

High-Quality Tutoring Works

50th → ~63rd

percentile: where a typical tutored student ends up

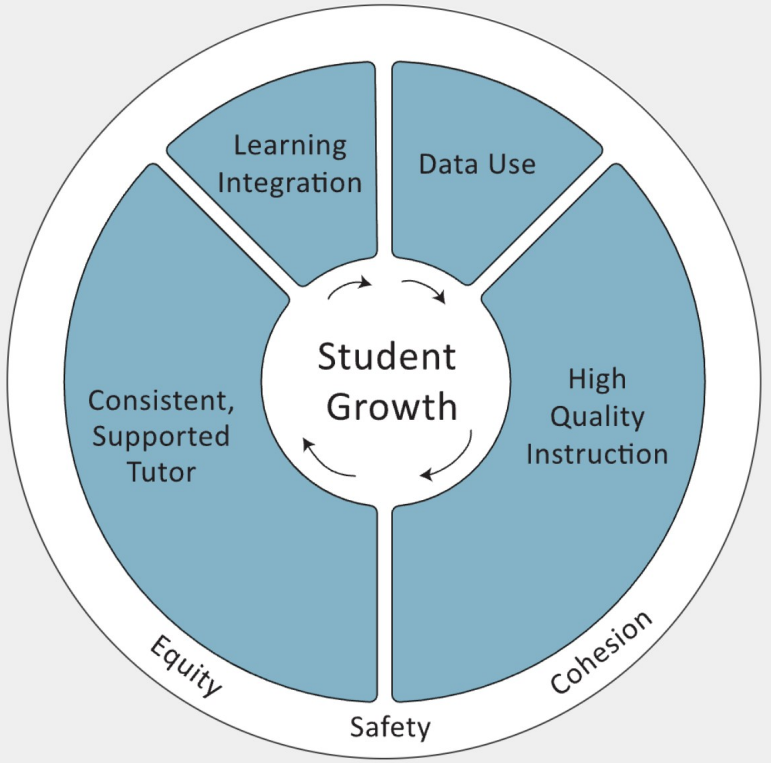
- Meets individual needs
- Builds relationships
- Supports well-being
- Increases attendance
- Accelerates learning by 3–15 months

Stanford SCALE / National Student Support Accelerator · Loeb & Robinson 2026 synthesis of meta-analyses

TOPIC 1 · STUDYING HELP

What the Best Tutoring Programs Share

1	Frequent sessions	AI: strong
2	Aligned with the course	AI: weak
3	Uses evidence to adjust	AI: partial
4	The same tutor over time	AI: partial
5	Trained, supported tutors	AI: weak



The Elements of High-Impact Tutoring

Stanford SCALE, “AI Tutoring Is Not a Monolith” and “Homework Helper or AI Tutor?” (Aug 2026)

DISCUSSION

Can the friend close these gaps?

Aligned with the course | Uses evidence to adjust | The same tutor over time | Trained, supported tutors

1

What would it take for AI to do this well?

2

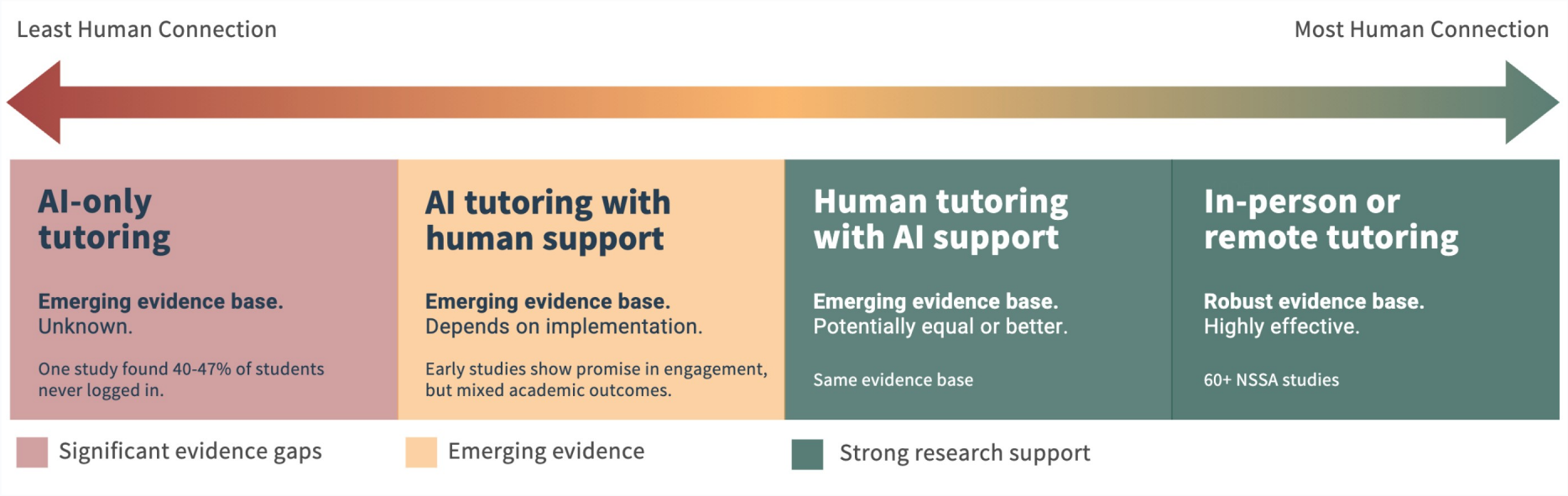
Which can it never do well, however it's set up? Why?

3

What requires a human connection?

TOPIC 1 · STUDYING HELP

A Spectrum, From All AI to All Human



The human end works, but it's hard to scale

The AI end scales, but doesn't always work

Stanford SCALE / National Student Support Accelerator, "AI Tutoring Is Not a Monolith" (Aug 2026)

TOPIC 1 · THE AI-ONLY END

AI Alone: Better Homework, Worse Exams

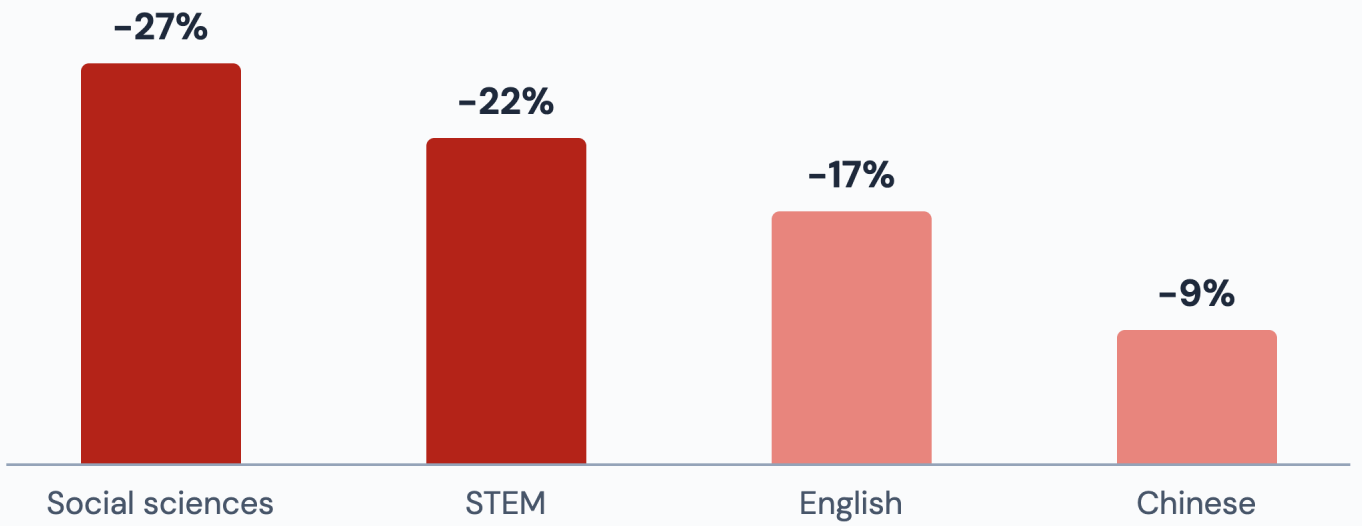
+18%
homework scores

−20%
monthly exam scores

Homework time fell from 64 to 45 minutes.

The losses grew over two years: high school entrance exams **−24%**, college entrance exams **−18%**.

Drop in exam scores after starting AI, by subject



Strömberg, Lei & Wu, CEPR Discussion Paper 21577 (VoxEU, Sep 2026) · 26,811 students, grades 7–12, central China, 30 months

TOPIC 1 · IMPLEMENTATION

The Design Details Decide the Result

What the AI gives

Helped: one step or hint at a time (Harvard)

Hurt: full answers (Bastani: exam −17%)

Who tries first

Helped: student attempts before help (Harvard)

Hurt: AI first, weaker engagement (Kosmyna)

What it knows

Helped: expert solutions and a designed sequence (Harvard)

No effect: course materials alone (Thoeni & Fryer)

How it's offered

Helped: scheduled, supervised sessions (Nigeria, Ghana)

Hurt: just made available (Maryland: −0.37 SD)

Who it talks to

Helped: coaching the human tutor (Tutor CoPilot)

Weak: students alone (40–47% never logged in)

Same underlying models, very different outcomes.

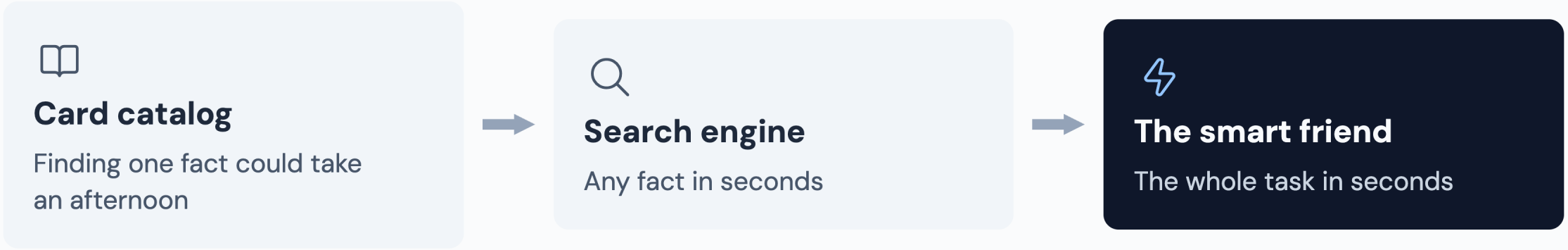
TOPIC 2

Learning Outcomes

What should students leave our courses able to do themselves?

TOPIC 2 · LEARNING OUTCOMES

When the Answer Is Seconds Away



What do we still want students to be able to do **themselves**, and why?

TOPIC 2 · PAIR DISCUSSION

Pair Up Across Departments

Pick a course you teach regularly. Take turns explaining:

1

The big picture

What do you hope students leave with, years from now?

2

Two or three concrete skills

What should they be able to do, and why does it matter in your field?

3

Why do it themselves?

For each skill: what's lost if they just ask the friend?

Then we'll share with the whole group.

TOPIC 3

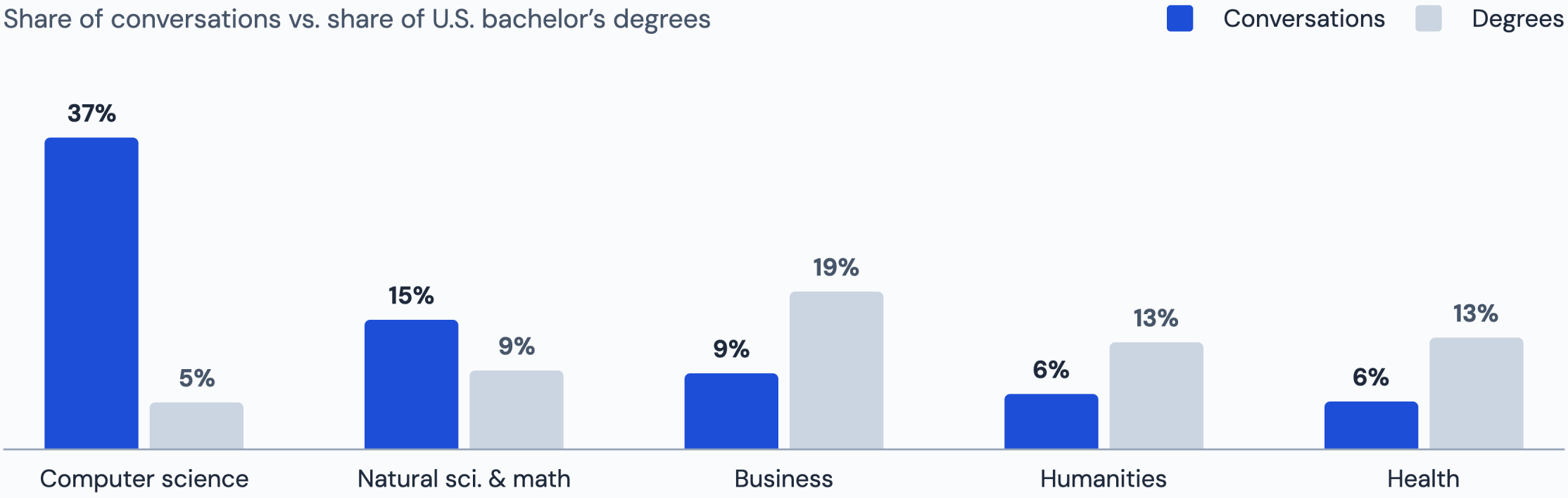
Students, AI and Detectors

How students actually use the friend, and whether we should police it

TOPIC 3 · HOW STUDENTS USE AI

574,740 Student Conversations

Share of conversations vs. share of U.S. bachelor's degrees



Anthropic Education Report: How University Students Use Claude (April 2025) · 18 days of Claude.ai conversations from higher-ed accounts

TOPIC 3 · HOW STUDENTS USE AI

About Half Just Ask for the Answer

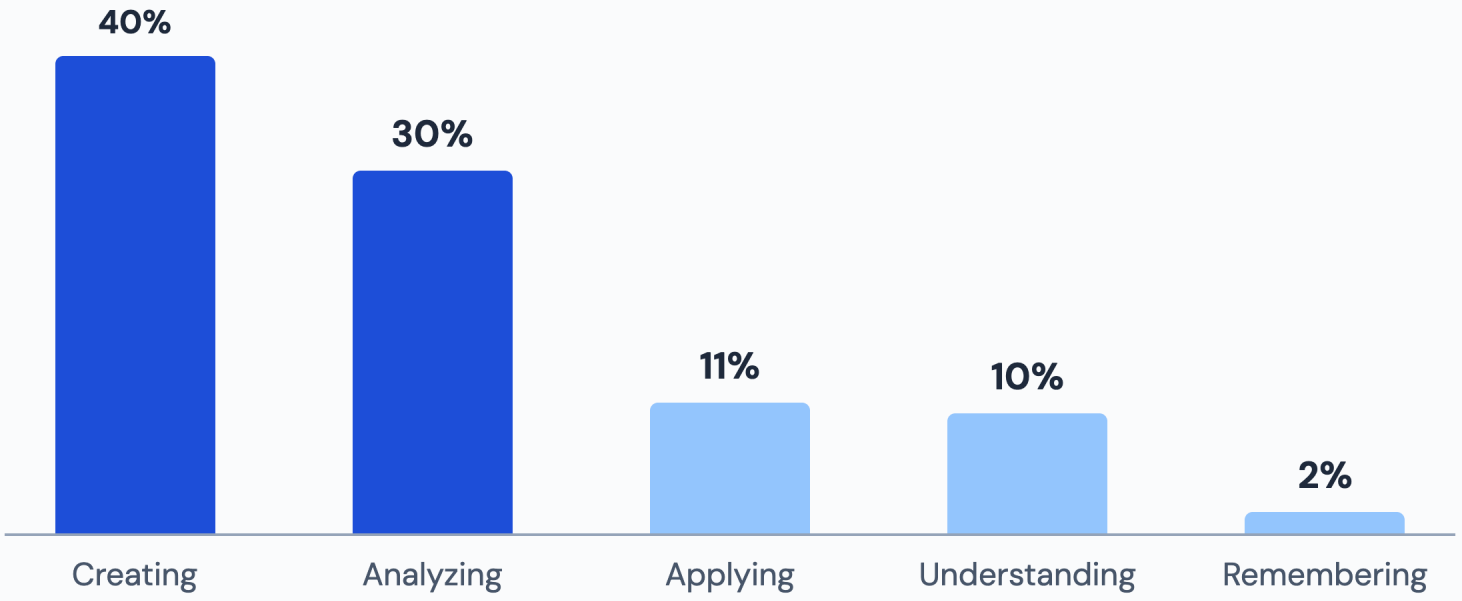


Anthropic Education Report (April 2025) · quoted requests are the report's summaries of conversations

TOPIC 3 · HOW STUDENTS USE AI

AI Does the Higher-Order Thinking

What Claude was doing, by level of Bloom's taxonomy



The skills we most want students to build are the ones they hand off most.

Anthropic Education Report (April 2025) · Bloom's level of the task Claude performed

DISCUSSION

How should students use AI in your course?

1

What would you want them to ask it for?

2

What should they never ask them for, and why?

3

How would you tell them, and would they listen?

TOPIC 3 · AI DETECTORS

2023–24: Detectors Didn't Work

61%

of essays by non-native English writers flagged as AI

Liang et al. 2023 · 7 detectors

14 tools

"neither accurate nor reliable"

Weber-Wulff et al. 2023 · incl. Turnitin

40% → 22%

accuracy after simple edits; 15% of human texts flagged

Perkins et al. 2024 · 7 detectors

Liang et al., Patterns 2023 · Weber-Wulff et al., Int. J. Educational Integrity 2023 · Perkins, Roe et al., IJETHE 2024

TOPIC 3 · AI DETECTORS

2025+: The Best Detectors Are Very Accurate

False positive rates of AI detectors, by text type

	Pangram	GPTZero	Originality.ai
Product review	0.5%	2.4%	1.7%
Blog post	0%	0.5%	0%
News article	0.1%	1%	0.3%
Novel excerpt	0%	0.5%	0.3%
Restaurant review	0.8%	1%	2.2%
Résumé	0%	0%	0.1%

Jabarian and Imas, 2025

≈2% or less

of AI text missed, all three tools

After a "humanizer" rewrite, missed: Pangram 0–5%, GPTZero 26–77%

Jabarian & Imas, NBER Working Paper 34223 (2025) · ~2,000 human passages vs. text from GPT-4.1, Claude Opus 4 and Sonnet 4, Gemini 2.0 Flash

TOPIC 3 · AI DETECTORS

Small Error Rates Add Up

An honest student submits 40 papers over four years. Chance of being flagged at least once:

1% false positives

33%

0.5% false positives

18%

0.1% false positives

4%

P(flagged at least once) = $1 - (1 - p)^{40}$ · assumes independent checks

DISCUSSION

Do you want to be the AI police?

A detector says a student's paper is 98% likely AI. The student says they wrote it.

What happens next, step by step?

Is that a process you want to run, all semester?

Or would you rather redesign the assignment?